The Effect of *Garcinia Kola* on Some Serum Oxidative Stress and Anabolic-Catabolic Biological Markers in Male Wistar Rats

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ABSTRACT

This study investigates the effect of *Garcinia kola* (*G. kola*) seed on some serum oxidative stress and anabolic-catabolic biological markers in rat model. Daily varying doses of crude *G. kola* seed (1000mg/kg (group B), 1200mg/kg (group C) and 1400mg/kg (group D)) was mixed with feed of rats (180-200g) while the control (group A) had feed only. At the end of seven days on these diets, blood sample was obtained for estimation of serum protein, Malondialdehyde-MDA, Superoxide dismutase-SOD, and Catalase-CAT, testosterone and cortisol. The data were subjected to statistical analysis (ANOVA) and LSD post-hoc test at $p<0.05$ using SPSS (version 20). The results revealed increased body weight ($p>0.05$), dose dependent increased in serum CAT (220.04±21.19; 342.29±29.95; 456.48±31.33l and 485.04±32.47 U/mg protein), SOD (17.00±2.17; 20.19±2.17; 26.24±1.59; 33.67±3.85 U/mg protein) and testosterone (4.12±0.77; 5.52±1.12; 5.96±01.07; 6.48±1.35 ng/ml) by graded doses of *G. kola*. On the other hand, there was dose dependent non-significant decrease ($p>0.05$) in serum MDA (4.77±0.47, 4.02±0.55, 3.62±0.64 and 3.01±0.15 U/ml) and protein (3.45±0.31, 2.90±0.95, 2.83±0.41 and 2.17±0.07 mg/ml) levels by graded doses of *G. kola*. There was dose dependent significant increase in the testosterone-cortisol ratio in the 1200mg/kg (0.63±0.07) and 1400mg/kg (0.76±0.04) treated groups compared with the untreated control (0.41±0.07). Overall, the anabolic action of *G. kola* indicated by weight gain is likely via its improvement of serum testosterone and antioxidants and mitigation of oxidative stress products in normal rats.

Keywords: Garcinia kola/ bitter kola, anabolic-catabolic balance, oxidative stress, antioxidants.

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1. INTRODUCTION

Medicinal plants and products have gained popularity and there is now increased utilization world-wide. *Garcinia kola* seed (*G. kola*) or "bitter kola" as it is called because of its bitter taste, is one that has enjoys folk reputation in Africa for its aphrodisiac property[1]. Among the Nigerian languages, it is called "Miji-goro", "Akilu or Ugolo" and "Orogbo" by the Northern, Eastern and Western people of Nigeria respectively. It has a long history for the Africa traditional healer as it is used as food preservative, poison antidote[2], treatment of diarrhoea[3], throat and bronchi infections[4], cough, purgative, parasitic, microbial and viral infections[5-8]. Studies have also documented its used in the management of liver disorders[3, 9] and sickle cell disease[10]. Other studies have shown its anti-hepatotoxic[3, 9], anti-inflammatory[11], analgesic and hypoglycemic properties[12, 13]. *G. kola* has been reported as a good source of natural antioxidant[14, 15] and to contain phyto-chemical such as phenol, steroids, xanthines, benzophenones, tannins, guttiferins and saponins, biflavonones and kola-flavonone[16].

In mammals, including humans; growth is a normal physiological process and is influenced by what is ingested. The process of growth is accompanied by characteristic changes in energy balance[17] which is determined by the anabolic and catabolic balance; a marker of oxidative stress[18] systemic inflammation[19]. Testosterone possesses anabolic effects while cortisol catabolic effects and thus, the testosterone/cortisol ratio have been recognized as a good marker for stress and metabolism[20]. The need to examine the effect of this medicinal plant on energy balance cannot be overemphasized. Although a number of peptide hormones had been shown

to influence the components of energy balance (food intake, metabolic rate, body weight, body temperature, etc.) by having either an overall anabolic or catabolic effect[21], the effect of medicinal plants like *G. kola* has not been well investigated. Despite its traditional uses, beneficial experimental findings and rich nutritional content, its effect on oxidative stress status of the general biological system are still been debated and its metabolic influence on testosterone-cortisol response have not been extensively investigated. The presence of xanthone and biflavonoids in the seed[22. 23] and the documentation that the seed contain antioxidant potentials to improve oxidative status[24, 25] justifies evaluation of the plant. Thus, the present study investigates the effect of graded doses of *G. kola* on some serum biological stress and testosterone-cortisol ratio in adult male Wistar rats.

2. MATERIALS AND METHODS

Adult male Wistar rats (N=20) of comparable weight (180 to 200g) were obtained from Animal holding facility of the Department of Physiology, Ambrose Alli University, Ekpoma. They were transferred to the Physiology laboratory II where the experiments were conducted. The animals were housed in plastic cages (50cm x 35cm x 25cm) under 12/12hr day light/ night dark cycle and allowed 14 days to acclimatize to the laboratory conditions before the experiments. They were fed *ad libitum* with rat chow and clean water during acclimatization and the experiment. Approval for the study was obtained from the Ambrose Alli University Health Research Ethics Committee (assigned number: 34/18) and was conducted following Guidelines for Care and Use of Laboratory Animals in Biomedical Research

Garcinia kola (*G. kola*) seeds were purchased commercially from Ekpoma, Edo State, Nigeria. some of the seeds were taken for authentication in the herbarium, Forestry Research Institute, Ibadan, Nigeria (FRI: 110252). After authentication, the testas of the seeds were peeled and the seeds were cut into small sizes and sun dried. The dried smaller sizes seeds were blended into powder, packed in clean containers until when used.

Based on the results of the oral acute toxicity[26], the 20 rats were divided into 4 groups with group 1 as the untreated control and groups 2, 3 and 4 as the test groups treated with 1000mg/kg, 1200mg/kg and 1400mg/kg dose of the powder *G. kola* seed respectively. Administration of the *G. kola* seed was as describe by Nwaopara *et al.*[27] and previously documented by Osifo and Iyawe[26].

Body weight was determined prior to treatment and after 7 days of the various treatments. Thereafter, the animals were euthanized via cervical decapitation and blood was collected through the jugular vein into plain sample bottles and allowed to clot. The clotted blood was centrifuge at 4000rpm for 10minutes and the serum decanted into new sample bottle and freezed until when assays were done.

The serum protein concentration was determined using Biuret method as documented by Gornall *et al.*[28] while lipid peroxidation was determined by measuring the thiobarbituric acid reactive substances (TBARS) produced during lipid peroxidation by the method of Varshney and Kale [29]. The serum levels of SOD and Catalase were determined by the methods of Misra and Fridovich[30] and Sinha[31] respectively.

The data were analyzed using ANOVA in SPSS 20 and $p<0.05$ was determined using LSD posthoc analysis. The results are presented as means ± SEM in bar chart and tables.

3. RESULTS

Figure 1 shows the effect of graded doses of *G. kola* treatments on body weight changes. Treatments with varying doses of *G. kola* (184.80±8.66g vs 191.08±18.51g;

184.28±7.60g vs 195.66±11.74g and 187.12±5.56g vs 197.56±14.08g for 1000mg/kg, 1200mg/kg and 1400mg/kg G.k treated respectively) cause a non-significant increase ($p>0.05$) in the final mean body weight compared to the untreated-control (185.00±3.45g vs 189.68±10.19g).

Table 1 shows the effect of graded doses of *G. kola* on some serum oxidative and antioxidant variables. Treatment with varying doses of *G. kola* causes a dose dependent non-significant decrease ($p>0.05$) in the levels of serum MDA (4.77±0.47 U/ml, 4.02±0.55U/ml, 3.62±0.64 U/ml and 3.01±0.15 U/ml for Control, 1000mg/kg, 1200mg/kg and 1400mg/kg *G. kola* treatments respectively) and protein (3.45±0.31 mg/ml, 2.90±0.95mg/ml, 2.83±0.41 mg/ml and 2.17±0.07 mg/ml for Control, 1000mg/kg, 1200mg/kg and 1400mg/kg *G. kola* treated respectively). However, there was a dose dependent significant increase ($p<0.05$) in serum CAT activity (220.04±21.19 U/mg protein, 342.29±29.95 U/mg protein, 456.48±31.33 U/mg protein and 485.04±32.47 U/mg protein for control, 1000mg/kg, 1200mg/kg and 1400mg/kg *G. kola* treated respectively) following *G. kola* treatments compared with the control. On the other hand, *G. kola* treatments cause a dose dependent increase in serum SOD activity that is not significant with 1000mg/kg (20.19±2.17U/mg protein) and 1200 mg/kg (26.24±1.59 U/mg protein) but significant ($p<0.05$) with 1400mg/kg (33.67±3.85 U/mg protein) *G. kola* treatment compared with the control (17.00±2.17 U/mg protein).

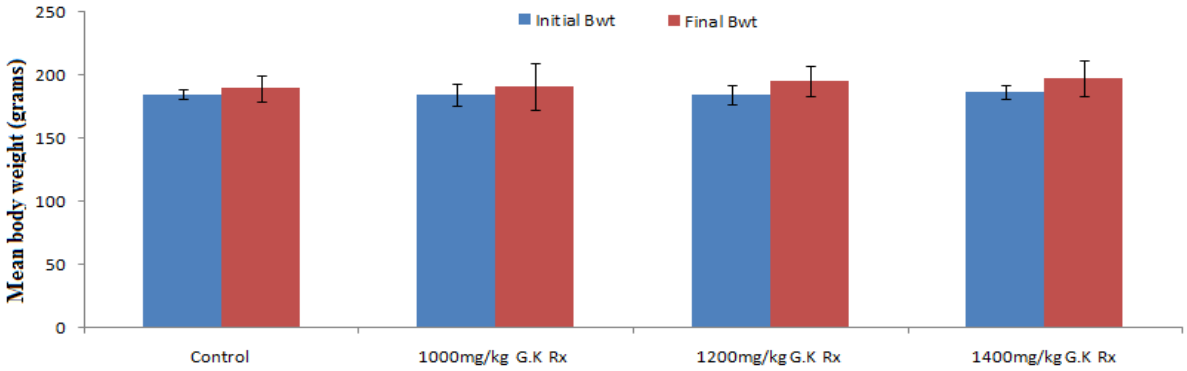


Figure 1. The effect of graded doses of *G. kola* treatments on initial and final body weights
Values are mean ± SEM, N= 20 male rats, n= 5 male rats.

Table 1. Serum oxidative stress and antioxidant marker activities following graded doses of *G. kola* treatments

Groups	Serum oxidative stress markers		Serum antioxidant marker	
	Serum MDA (U/ml)	Serum proteins (mg/ml)	Serum SOD (U/mg protein)	Serum CAT (U/mg protein)
Control	4.77±0.47	3.45±0.31	17.00±2.17	220.04±21.19
1000mg/kg *G.K* Rx	4.02±0.55	2.90±0.95	20.19±2.17	342.29±29.95*
1200mg/kg *G.K* Rx	3.62±0.64	2.83±0.41	26.24±1.59	456.48±31.33*
1400mg/kg *G.K* Rx	3.01±0.15	2.17±0.07	33.67±3.85*	485.04±32.47*

Values are in Mean ± SEM, N = 20 male rats, n= 5 male rats, *significantly different at $p<0.05$ compared to control, MDA = Malondialdehyde, SOD = Superoxide dismutase, CAT = Catalase.

Table 2 presents the effect of graded dose treatments of *G. kola* on serum testosterone, cortisol and testosterone-cortisol ratio. Serum testosterone level was non-significantly higher ($p>0.05$) in the 1000mg/Kg *G. kola* treated group (5.52±1.12ng/ml) but significantly increase ($p<0.05$) in the 1200mg/kg (5.96±01.07 ng/ml) and 1400mg/kg (6.48±1.35 ng/ml) *G. kola* treated groups compared with the untreated-control (4.12±0.77 ng/ml). On the other hand, *G. kola* treatments resulted in dose dependent decreased in serum cortisol levels; but was only significant in the 1400mg *G. kola* treated group (8.57±1.46 ng/ml), compared with the control (10.06±1.44 ng/ml). *G. kola* treatments caused a dose dependent increase in the testosterone-cortisol ratio compared with the untreated control (0.41±0.07) and were significant in the 1200mg/kh (0.63±0.07) and 1400mg/kg (0.76±0.04) treated group.

Table 2. Serum testosterone and cortisol activities and testosterone-cortisol ratio following graded doses of *G. kola* treatments

Groups	Serum testosterone (ng/ml)	Serum cortisol (ng/ml)	Testosterone-cortisol ratio
Control	4.12±0.77	10.06±1.44	0.41±0.07
1000mg/kg *G.K* Rx	5.52±1.12	9.97±0.98	0.55±0.09
1200mg/kg *G.K* Rx	5.96±1.07*	9.42±1.35	0.63±0.07*
1400mg/kg *G.K* Rx	6.48±1.35*	8.57±1.46*	0.76±0.04*

Values are in Mean ± SEM, N = 20 male rats, n= 5 male rats, *significantly different at $p<0.05$ compared to control.

4. DISCUSSION

Biological system generation of reactive oxygen compounds is inevitable since they are product of metabolic processes. However, reactive oxygen species are considered biologically toxic because they damage proteins, nucleic acids and lipids[32, 33]. To avoid this damaging effect, cells employed collections of antioxidants including antioxidant enzymes, vitamins and thiols, to neutralize the generated oxygen species[32, 34]. Interestingly, plant extracts and products for enhancing organs and body systems performance in man and animals have gained increasing acceptance[35]. In the present study, varying dose ingestion of *G. kola* in normal rats was observed to enhance body weight, and the levels of serum CAT, SOD, testosterone and testosterone-cortisol ratio while diminishing the levels of serum MDA, protein and cortisol. The observed findings that G, kola improved body weight disagrees with the results of Okorodudu *et al.*[36] *who* revealed non-statistically significant relative body weight loss in female rats exposed to 200 to 400 mg/kg body weight of aqueous extracts of *G. kola* for 21 days. The observed difference between the findings of the present study and that of Okorodudu *et al*. [36] suggests that prolonged intake may mildly reduce body weights. Also, considering the differences in the sex of the rats, sexual dimorphism in the metabolism of *G. kola* or the sex hormones may impact *G. kola* fate differently between genders.

In several situations, MDA, SOD and CAT activities can be used to understand the biochemical events of free radical damages and scavenging, however little is known whether stress is an important consequence of *G. kola* continuous intake[36]. In fact, studies had documented decreased oxidative stress and improved metabolic activities in diabetes mellitus following *G. kola*

treatment[37]. While this is an indication of ameliorating capacity of oxidative stress by *G. kola*, little is known of acute treatment of 7 days as several studies have been chronic. In this study, oxidative stress was studied by evaluation the serum levels of protein and MDA as well as serum antioxidant activities. The current study showed *G. kola* ingestion for 7 days ameliorates oxidative stress generated by normal metabolic processes while enhancing antioxidant level. In support of this observation in the present study, Ige *et al*.[38] revealed that *G. kola* extract prevented lipid peroxidation via stimulating the enzymatic antioxidants (catalase and superoxide dismutase) levels and reducing malondialdehyde (lipid peroxidation index).

On serum testosterone and cortisol, this study reveals *G. kola* present a negative correlating effect between the two hormones. Specifically, *G. kola* treatment caused an enhancing effect on serum testosterone but a weaken effect of serum cortisol. In line with the observed enhancement of serum testosterone, Akpantah *et al*.[39] and Braide *et al*.[40] have previously reported *G. kola* seed extracts administration to increase testosterone production in rats. Changes in sex hormones due to stress hormone (Cortisol) concentrations in sleep deprived animals fed with *G. kola* had been previously reported Hipólide *et al*.[41]. Considering testosterone as a possible anabolic hormone and cortisol as a catabolic one[20], the elevated level of testosterone by *G. kola* treatment indicate its anabolic action and thus the enhancement of anabolic catabolic ratio in this study.

5. CONCLUSION

Reactive oxygen species are considered to be unavoidable by-products of normal aerobic metabolism, a shift of the anabolic-catabolic homeostasis to the catabolic side can disrupt cellular homeostasis. In our study, *G. kola* treatment was observed to shift the anabolic-catabolic balance to the anabolic side and thus the increased body weight. This action of *G. kola* was through enhancement of serum testosterone and antioxidants while alleviating the oxidative products of normal biological process.

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